



Ganga Saathi Circles

A Community-Driven Waste Segregation Policy for Varanasi Slums

Focus Area

Urban Sustainability

Problem Statement

Waste Segregation in Varanasi Slums

Event

EUREKA – INNORAVE 2026

The Unseen Crisis

Why Waste Segregation is Failing in Varanasi

Varanasi, a city rich in cultural heritage, faces a severe challenge in waste management. Annually, it generates an astounding 1,200 tons per day (TPD) of waste, yet a mere 12% is segregated at the source. This critical failure impacts both environmental health and public well-being.

Slums situated near prominent institutions like IIT-BHU disproportionately contribute to this problem, generating 25% of the city's unsegregated waste. This direct dumping of mixed waste contaminates the sacred Ganga river, spreads disease, and significantly increases municipal cleanup costs, creating a continuous cycle of environmental degradation and public health risks.

Core Issue: While infrastructure and awareness exist, the crucial elements of **incentives and accountability are missing** at the community level.

The Human Element of Waste

What We Discovered: A Community-Centric Insight

"People do not segregate waste for the city – they do it when their family and community benefit."

Our research reveals a fundamental disconnect: traditional top-down waste management approaches overlook intrinsic human motivations. We found that individuals are less likely to adopt segregation practices when there's no direct, tangible benefit to their immediate household or community.



No Household-Level Reward

The effort of segregating waste often goes unacknowledged, offering no direct financial or social return to the individual household.



No Community Ownership

Without a shared stake or collective goal, the responsibility for waste management feels distant and institutional, not personal.



No Visible Economic Value of Waste

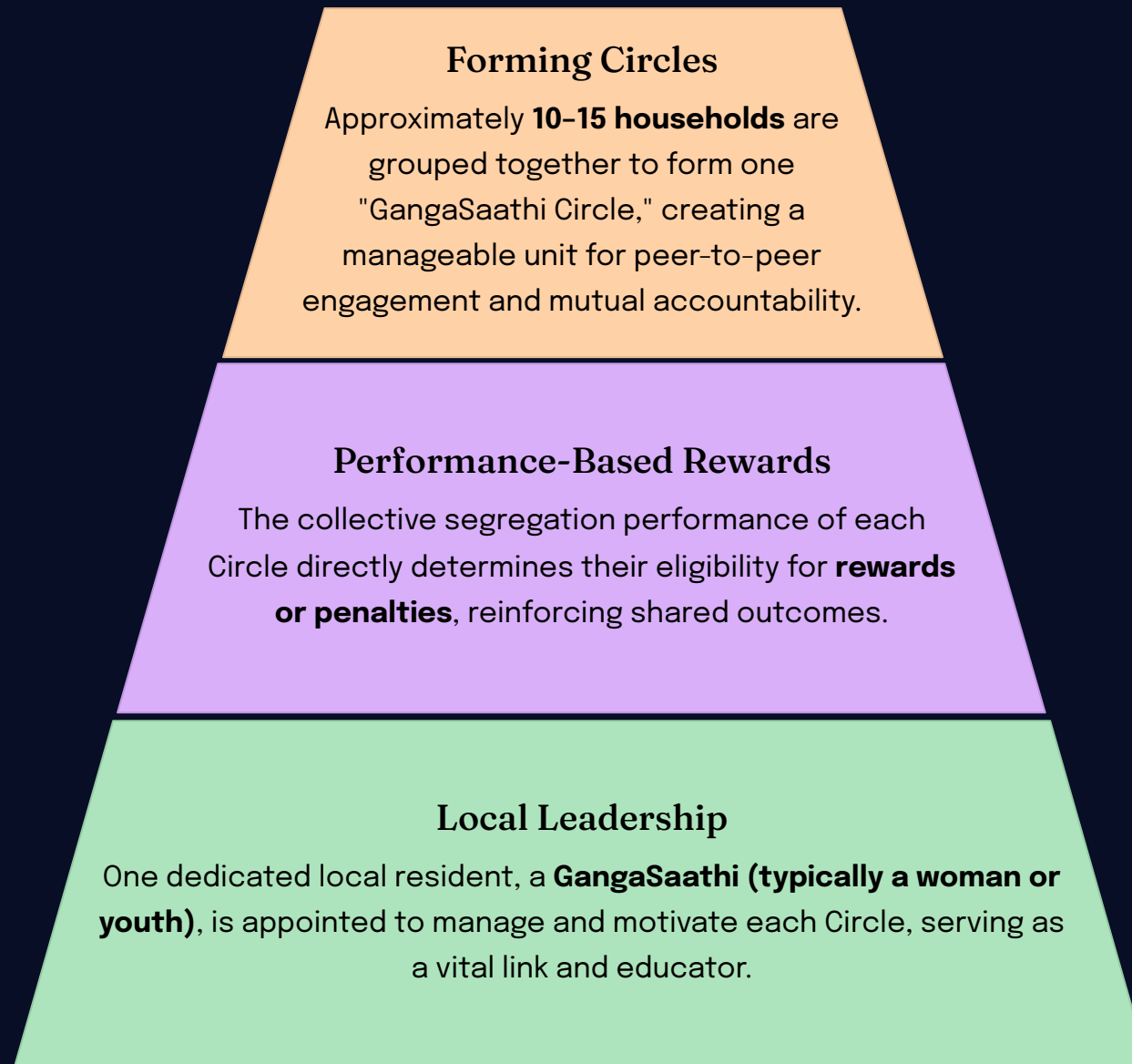
The potential economic benefits of sorted recyclables are not realized or channeled back into the community, perpetuating the view of waste as valueless.

→ **The solution must integrate social and economic drivers, extending beyond mere regulatory enforcement.**

Introducing GangaSaathi Circles

Transforming Waste into a Community Asset

GangaSaathi Circles proposes a revolutionary, community-centric model that harnesses local social dynamics and economic incentives to drive sustainable waste segregation. This initiative aims to foster a sense of collective responsibility and empower local residents.



This approach transforms waste segregation from a municipal burden into:

- **A Community Habit:** Integrating segregation into daily life through social norms.
- **A Source of Income:** Creating new economic opportunities and tangible benefits.
- **A Shared Responsibility:** Fostering collective ownership and pride in a clean environment.

The Operational Blueprint

How the GangaSaathi Policy Works



Mandatory 3-Bin System

Each household will be provided with a set of three color-coded bins to simplify waste sorting:

- **Green Bin:** Exclusively for wet biodegradable waste, destined for composting.
- **Blue Bin:** For dry recyclable materials, ensuring they can be processed efficiently.
- **Red Bin:** Dedicated to sanitary and hazardous waste, for safe disposal.



Operational Model

A structured daily routine ensures compliance and efficiency:

- **Daily Segregation Check:** GangaSaathis perform regular checks to ensure proper sorting.
- **Door-to-Door Separate Collection:** Waste is collected daily, keeping segregated streams intact.
- **Wet Waste to Compost:** All wet waste is transported to local composting units.
- **Dry Waste to Recyclers:** Recyclables are channeled directly to authorized recycling facilities.

① This policy leverages and integrates seamlessly with **existing Swachh Bharat Mission (SBM) framework**, requiring no new overarching laws, only localized implementation protocols.

Driving Change Through Incentives

Behaviour Change & Accountability Model

To foster consistent participation, the GangaSaathi Circles policy implements a transparent incentive and penalty system. This model directly links household segregation efforts to tangible benefits, ensuring immediate and visible returns on their commitment.

90%+ compliance

₹300 / household

75–90% compliance

₹150 / household

<75% compliance

Warning → small fine

These incentives are carefully chosen to directly impact the daily lives and well-being of the participating households, making the benefits of segregation immediately apparent:

- Grocery vouchers for essential household items.
- Mobile recharge credits, a vital utility in modern India.
- School supplies for children, supporting education.
- Rebates on electricity bills, reducing household expenses.

→ This direct feedback loop creates immediate, visible benefits, reinforcing positive behavior and community engagement.

A Collaborative Ecosystem

Key Stakeholders and Their Vital Roles

The success of GangaSaathi Circles hinges on a robust partnership between various entities, each playing a crucial role in the waste management ecosystem. This collaborative approach ensures accountability, efficiency, and sustainability across all levels.



This multi-stakeholder model not only ensures effective waste management but also **creates local jobs and fosters trust-based governance** within the communities.

Investing in a Cleaner Future

Financial Feasibility: Annual Pilot Cost (1,000 Households)

The GangaSaathi Circles pilot program for 1,000 households is designed to be highly cost-effective, especially when compared to the long-term environmental and health costs associated with unmanaged waste. Our projections demonstrate a prudent investment with significant returns.

Bins & Signage	12 lakhs
Incentives	18 lakhs
GangaSaathi Honorarium	6 lakhs
Training & Monitoring	4 lakhs
Total Annual Pilot Cost	₹40 lakhs

This budget allocation prioritizes tangible support for households and local leaders, directly funding the core mechanisms of behavior change and operational efficiency.

 The annual pilot cost of **₹40 lakhs** is a modest investment compared to the enormous long-term expenses incurred from river cleanup, public health interventions, and environmental remediation necessitated by poor waste management.

Transformative Impact & Scalability

Measurable Outcomes and Future Expansion

The GangaSaathi Circles initiative promises significant, measurable improvements within a short timeframe, demonstrating its efficacy and potential for widespread adoption. Our 12-month projections highlight substantial positive shifts in waste management practices and community well-being.

1

Source Segregation

Projected increase from 12% to 70% in source segregation within 12 months.

2

Ganga Pollution

Expected reduction of waste entering the Ganga river by half.

3

Ragpicker Income

Anticipated income increase for ragpickers due to higher quality segregated waste.

4

Community Participation

Expected engagement rate among households participating in the program.

Scalability: A Blueprint for National Impact

- **Pilot to City-Wide:** Successful pilot results will inform expansion to all slums across Varanasi.
- **Inter-City Expansion:** The model is designed for replication in other Ganga river cities facing similar waste management challenges.
- **National Replicability:** The framework is fully compatible and replicable under the existing **Swachh Bharat Mission**, offering a proven solution for urban sustainability nationwide.

Mitigating Risks, Maximizing Impact

Addressing Challenges and Concluding Vision

While any new policy implementation carries inherent risks, GangaSaathi Circles has proactive mitigation strategies in place to ensure smooth operation and long-term success, built on community engagement and transparent governance.

Community Resistance	Initiate with women-led circles and local champions to build trust and ownership.
Incentive Misuse	Implement digital tracking and regular audits to ensure fair and transparent distribution of benefits.
Funding Gaps	Diversify funding through sustained CSR partnerships and municipal convergence with existing budgets.

Conclusion: A Community-Owned Solution

GangaSaathi Circles converts waste segregation from a municipal burden into a community-owned, incentive-driven system that protects the Ganga and improves lives. By fostering local leadership, providing tangible benefits, and empowering households, this policy creates a sustainable model for urban waste management that can be replicated across India.